Parameters Needed in the Pricing Model

Symbol	Description
S_t	Price of the Risky Asset at time t.
B_t	Price of the Risk-Free Asset at time t.
μ	Drift rate of S_t.
σ	Volatility of S_t.
r	Risk-free interest rate.
θ	Bid-ask spread factor: $\theta = \frac{S_{\text{ask}} - S_{\text{bid}}}{2S}$.
S_{ask}	Ask price: $S_{\text{ask}} = (1+\theta)S$.
S_{bid}	Bid price: $S_{\text{bid}} = (1-\theta)S$.
α_t	Shares account (number of risky shares held).
β_t	Money account (investment in risk-free asset).
L_t	Cumulative purchase of risky assets up to t.
M_t	Cumulative sale of risky assets up to t.
W_t	Liquid wealth: $W_t = \beta_t + S_t(\alpha_t - \theta\|\alpha_t\|)$.
δ	Derivative position: $\delta = -1$ (short), $\delta = +1$ (long).
$C_T(S)$	Terminal payoff of the derivative at T.
K	Strike price of the option.
U	Utility function of the investor.
$v_0(\alpha, \beta, s, t)$	Value function without derivative.
$v_\delta(\alpha, \beta, s, t)$	Value function with derivative position δ.
$z_0(\alpha, \beta, s, t)$	Certainty equivalent without derivative.
$z_\delta(\alpha, \beta, s, t)$	Certainty equivalent with derivative position δ.
V	Option price: $V = e^{-r(T-t)}(z_\delta - z_0)$.
t	Current time.
T	Terminal time (option maturity).
V_A, V_B, V_C	Differential operators in HJB equations.
w	Liquid wealth expression: $w = \beta + s(\alpha - \theta\|\alpha\|)$.
U^{-1}	Inverse utility function.

Table 1: Parameters needed in the Pricing Model

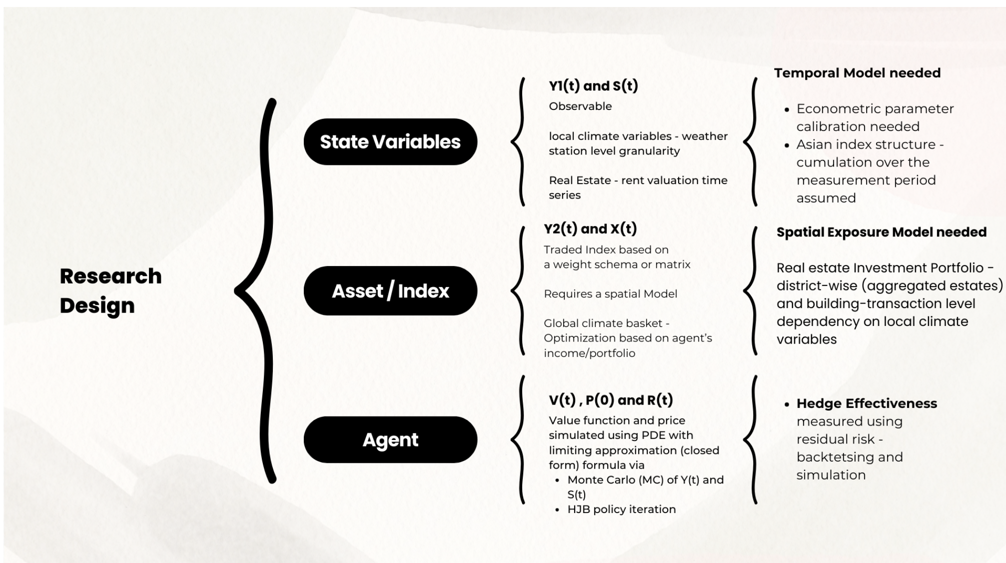

Figure 1: Research Design - Flowchart